

BEDR: Boundary Entropy-Driven Resampling with Large Language Models for Fine-Grained Cyber Threat Intelligence Mapping

Abstract

Automating the mapping of unstructured Cyber Threat Intelligence (CTI) to MITRE ATT&CK techniques (TTPs) is pivotal for proactive cyber defense. However, real-world datasets are plagued by extreme long-tail distributions—characterized by numerous zero-shot classes—and severe semantic overlap among fine-grained techniques. Traditional resampling methods, such as SMOTE, fail to address the “cold start” dilemma in zero-shot scenarios, while naive data augmentation strategies using Large Language Models (LLMs) are often compromised by mode collapse and semantic noise. To bridge these gaps, this paper proposes the BEDR (Boundary Entropy-Driven Resampling) framework. Unlike previous methods, BEDR aims to construct a comprehensive Atomic Action Benchmark Dataset covering all 679 MITRE ATT&CK techniques. It utilizes Boundary Entropy (H_{bd}) to provide a theoretical analysis of decision boundary ambiguity. The framework ensures high-quality sample generation for 105 zero-shot classes, effectively resolving the “cold start” dilemma. To ensure rigorous evaluation, we validated BEDR on a strictly independent test set collected from in-the-wild sources (e.g., security blogs and vulnerability reports), which mirrors the natural long-tail distribution of real-world threats. Results show that BEDR achieves a 22.3% improvement in Macro-F1, demonstrating its superior capability in mapping fine-grained TTPs.

Keywords: Cyber Threat Intelligence, TTPs, Imbalanced Learning, Large Language Model, Data Augmentation, Boundary Entropy

1. Introduction

Tactics, techniques and procedures (TTPs) are of paramount importance for guiding cybersecurity defense and characterizing cyberattack processes. The task of Cyber Threat Intelligence (CTI) mapping involves classifying unstructured texts—such as threat reports and blogs—into corresponding techniques based on the MITRE ATT&CK framework. MITRE ATT&CK (v17.1) defines 679 technical categories, comprising 211 parent techniques and 468 sub-techniques. Each category is accompanied by a description, and most are annotated with procedure examples. Typically, these descriptions and procedure examples serve as the foundational samples for training classification models.

However, datasets constructed in this manner suffer from severe class imbalance (manifesting as extreme long-tail distributions and numerous zero-shot classes) and high inter-class overlap. Specifically, there is a significant disparity in sample quantities: major classes may contain up to 480 samples,

whereas minor classes often have zero. Furthermore, the inherent inheritance relationship between parent and sub-techniques results in substantial semantic overlap, posing immense challenges for classification.

Existing research primarily addresses dataset balance by actively discarding small-sample classes during dataset construction. Regarding class overlap, there is a lack of dedicated discussion; a prevalent practice is to ignore sub-techniques or merge them into their parent categories. For instance, T1548 (Abuse Elevation Control Mechanism) is a parent technique containing six sub-techniques (e.g., T1548.001 Setuid and Setgid, T1548.002 Bypass User Account Control, etc.). The common approach consolidates these distinct sub-techniques under the single label T1548. While this avoids the overlap issue between parent and child categories and increases sample size, it introduces significant drawbacks.

Although such consolidation may improve macro-level classification accuracy, it inevitably leads to

several critical issues: (1) discarding small-sample classes renders the model impractical due to a lack of recognition capability for rare threats; (2) merging parent-child categories results in the holistic abandonment of 468 sub-techniques in ATT&CK, forcibly smoothing out semantic distinctions and thereby weakening the granularity of threat characterization; (3) forcing sub-techniques into parent categories may mitigate overlap definitions but exacerbates confusion at decision boundaries. This is manifested as a dense aggregation of samples near the parent category boundary in high-dimensional space, further elevating classification error.

To address these deficiencies, this paper proposes a Boundary-Entropy-Driven Resampling (BEDR) framework, aimed at simultaneously resolving class imbalance and boundary overlap without sacrificing granular detail. The core philosophy of this framework is a paradigm shift from passively accepting data defects to actively optimizing data quality. Rather than merely pursuing a balance in sample quantity, we focus on a more fundamental objective: reducing the ambiguity of decision boundaries.

To this end, our contributions are threefold:

- First, we formally define and introduce Boundary Entropy (H_{bd}) as a metric to quantify boundary ambiguity in TTP datasets. We theoretically demonstrate that reducing H_{bd} is an effective pathway to enhancing model generalization capabilities.
- Second, to overcome the limitations of standard embedding spaces, we innovatively construct a Hybrid Vector Scaffold (a combination of “Official Intent” and “Sentence Original Text”). This method is not used for final classification but serves as an advanced internal metric tool designed to calculate boundary entropy more reliably during the resampling process.
- Third, we design a multi-pass adaptive over-sampling algorithm. This algorithm utilizes Large Language Models (LLMs) to generate candidate samples and employs a Real-time Adaptive Threshold to intelligently screen them based on boundary entropy calculated via the aforementioned hybrid scaffold, achieving an optimal balance between sample quality and quantity.

The final output of the BEDR framework is an optimized text dataset. To verify its universal

value, we evaluate it within a standard pure sentence vector space. Experimental results demonstrate that models trained on our optimized dataset significantly outperform those trained on original datasets and other baseline methods across key metrics, fully validating the framework’s effectiveness in mitigating class imbalance and boundary overlap.

2. Related Work

2.1. Resampling Strategies for Imbalanced Learning

In addressing classification tasks with imbalanced data, data-level techniques Xie et al. (2020) are more commonly employed than algorithm-level methods Li et al. (2024b) Li et al. (2024a). These approaches primarily aim to mitigate class imbalance by balancing the sample distribution Chen et al. (2022). Among them, the Synthetic Minority Over-sampling Technique (SMOTE) Chawla et al. (2002) stands out as a representative algorithm. However, studies Koziarski (2020) Santos et al. (2022) indicate that poor classification performance is not solely a consequence of class imbalance; class overlap in real-world datasets often exacerbates the issue, posing a significant challenge in the field of imbalanced learning Azhar et al. (2022) Kumar et al. (2024).

Traditionally, class overlap is addressed by projecting data into a high-dimensional space using kernel functions to enhance linear separability Das et al. (2013). Despite this, the problem persists. Some studies suggest employing data cleaning techniques to manage overlap, such as Edited Nearest Neighbor (ENN) Wilson (2007), NearMiss-2 Mani and Zhang (2003), One-Sided Selection (OSS) Kubat (1997), SMOTE-ENN Batista et al. (2004), and SMOTE-IPF Sáez et al. (2015). However, these data cleaning strategies may lead to the loss of critical information and prove inadequate in certain scenarios.

In recent years, advanced resampling algorithms targeting boundary region samples, such as CHSmote Yuan et al. (2023), have emerged. Nevertheless, when class overlap is significant, these linear interpolation-based methods often introduce irrelevant boundary samples, thereby worsening the overlap. Some researchers attempt to circumvent this by assuming no initial overlap exists during resampling. For instance, the SyMProD algorithm

Kunakorn et al. (2020) calculates proximity coefficients between majority and minority classes, rejecting instances that exceed a specific threshold to avoid overlap. Tao et al. (2022) proposed a weight assignment strategy based on density and distance to form SVDD class boundaries, aiming to simultaneously address imbalance and detect post-generation overlap.

However, the reliability of these algorithms significantly declines when the original data inherently contains class overlap. To ensure the safety of generated samples, Shen et al. (2023) proposed the SSMOTE algorithm, which utilizes k-nearest neighbors (k-NN) to partition the feature space and sets a distance threshold to synthesize higher-quality samples with lower overlap within a stable feature space. Addressing overlapping datasets, Lee et al. (2018) used FSVM and KNN to divide the feature space into soft and hard overlap regions. They employed an OSM classifier and K-NN for classification; while effective, this approach increases the misclassification cost for non-overlapping samples and often overlooks the critical role of samples within the overlap regions.

To counteract the potential loss of minority samples during data cleaning, SMOTE-WENN Guan et al. (2021) considers feature space sparsity and enhances the SMOTE-ENN algorithm by adjusting distances between positive examples, preserving positive instances in overlap regions while removing negatives. To achieve clearer classification boundaries, Zhang et al. (2021) proposed the Random Forest Cleaning Rule (RFCL), identifying new marginal thresholds to reduce overlap. Additionally, P.V. et al. (2018) introduced the OBU Vuttipittayamongkol et al. (2018) and NBU Vuttipittayamongkol and Elyan (2020) algorithms. OBU utilizes membership values from the FCM algorithm to define overlap regions, while NBU extends detection through detailed recursive searches in neighborhood data. These algorithms highlight decision boundaries by removing negative examples from overlap zones. However, such practices significantly alter the original sample distribution and reduce the accuracy of negative (majority) classes, which is often undesirable in many practical applications.

In summary, the aforementioned studies primarily target class imbalance rather than class overlap. Furthermore, research focusing on class overlap often erroneously deletes a large number of majority samples in an attempt to accurately classify mi-

nority ones. Therefore, it is crucial to design an algorithm that simultaneously maintains the original sample distribution and effectively highlights decision boundaries for improving overall classification performance.

2.2. Oversampling Strategies Applied to CTI Mapping

Heejung Kim et al. (2022) identified that the primary obstacle in automated TTP classification is the quality of training data, specifically small dataset sizes and imbalance. Based on this, they proposed applying SMOTE Chawla et al. (2002) and EDA Wei and Zou (2019) techniques for data augmentation across 578 technical categories. Practical experiments demonstrated that augmenting the dataset to 115,030 samples using EDA yielded superior classification performance. Reference Rahman and Williams (2022) utilized SMOTE to expand 7,061 existing description sentences to 34,048 samples for 64 technical categories, achieving 87% precision and 84% recall. Reference Liu et al. (2022) proposed generating input embeddings using attack trigger words, Indicators of Compromise (IOCs Conti et al. (2018)), and description sentences. By fixing trigger words and IOCs while replacing other text content, they achieved data augmentation of 700 samples for each of the 7 technical categories to extract attack tactics. Reference Rani et al. (2024) utilized SecureBERT Aghaei et al. (2022) for data augmentation based on a masking mechanism. The model randomly masks tokens in base samples and selects the top-5 probable generated words to form new samples, retaining those that exceed a set cosine similarity threshold with the original sample. During training, utilizing SecureBERT and a linear classifier improved the recognition accuracy for 193 technical categories.

Although the above studies validate the feasibility of data augmentation in enhancing TTP classification to some extent, existing strategies exhibit significant limitations when facing real-world threat intelligence data characterized by high granularity and long-tail distributions.

First, whether it is SMOTE-like methods based on vector interpolation, or text augmentation methods based on rule perturbation (e.g., EDA) or local mask prediction (e.g., SecureBERT), they essentially belong to “passive quantitative balancing” strategies. They rely heavily on existing seed samples, rendering them helpless against the numerous

Zero-Shot classes in the ATT&CK framework, thus failing to solve the “cold start” dilemma.

Second, regarding sample generation quality, simple linear interpolation of vectors easily leads to synthetic samples detaching from the true semantic manifold, resulting in semantic noise. Meanwhile, rule-based token replacement, while preserving grammatical structure, struggles to introduce new attack scenarios and sentence variations, making the model prone to “Mode Collapse.”

Most critically, existing work generally lacks focus on the ambiguity of decision boundaries; blindly expanding samples in overlap regions often exacerbates inter-class confusion. In light of this, this paper proposes shifting the research paradigm from “quantity balancing” to “quality optimization.” We aim to address the zero-shot generation problem by introducing an external knowledge base (LLM) and actively optimizing decision boundaries using boundary entropy theory, thereby achieving precise mapping of fine-grained TTPs.

3. Methodology

The overall architecture of the proposed BEDR framework is illustrated in Figure 1. It primarily consists of three phases: structured semantic disentanglement, hybrid vector representation construction, and multi-pass adaptive resampling.

To facilitate the description of the proposed framework, the key mathematical notations and their corresponding definitions used throughout this paper are summarized in Table 1.

Table 1: Summary of Key Notations and Definitions

Notation	Definition
$\mathcal{D} = \{(x_i, y_i)\}$	The dataset containing samples x_i and labels y_i .
C	The total number of technique categories (679 in this study).
$\mathcal{N}_k(x)$	The set of k -nearest neighbors of sample x in the feature space.
$P(c \mathcal{N}_k(x))$	The posterior probability of class c in the local neighborhood of x .
$h(x)$	Local Boundary Entropy (LBE) of a specific sample x .
$H_{\text{gb}}(\mathcal{D})$	Global Boundary Entropy (GBE) of the entire dataset $\mathcal{D}$.
$G_c = [I, A, E]$	The structured attack graph for class c (Intent, Action, Entity).
v_{s_i}, v_{I_i}	The raw embeddings of the sentence s_i and the official intent I_i .
$\mathbf{z}_i$	The Hybrid Vector combining sentence semantics and official intent.
$\mathcal{Z}_{\text{ref}}$	The reference space constructed from hybrid vectors for entropy calculation.
ϵ_c	The adaptive boundary entropy filtering threshold for class c .
$\tau_{\text{maj}}, \tau_{\text{pass}}$	Thresholds for undersampling retention and oversampling targets.

3.1. Theoretical Derivation and Motivation of Boundary Entropy

3.1.1. Quantifying Decision Boundary Ambiguity

In fine-grained, high-overlap multi-classification tasks like CTI mapping, many samples do not reside

in clear, homogeneous regions within the feature space. Instead, they often locate in the “ambiguous zones” between multiple distinct categories—namely, the decision boundaries. A classifier is highly prone to errors in these regions. Consequently, a high-quality dataset should possess decision boundaries that are as distinct and definitive as possible. We require a metric to quantify the severity of such “boundary ambiguity.”

3.1.2. From Local Uncertainty to Local Boundary Entropy (LBE)

We begin with a single sample point x . The “boundary nature” of x can be characterized by the class purity of its local neighborhood. We employ the K-Nearest Neighbors (k-NN) concept to define this locality.

Local Posterior Probability: In a feature space $\mathcal{X}$, for any sample x , we observe its k nearest neighbors (excluding x itself), denoted as $\mathcal{N}_k(x)$. We estimate the posterior probability $P(c | \mathcal{N}_k(x))$ of class c appearing near x based on the labels y_j of these neighbors:

$$P(c | \mathcal{N}_k(x)) = \frac{1}{k} \sum_{j \in \mathcal{N}_k(x)} \mathbb{I}(y_j = c) \quad (1)$$

where C is the total number of classes (679 in this study), and $\mathbb{I}(\cdot)$ is the indicator function.

Local Boundary Entropy (LBE): The uncertainty of this probability distribution is naturally captured by Shannon Entropy. We define this value as the Local Boundary Entropy (LBE) of point x , denoted as $h(x)$:

$$h(x) := - \sum_{c=1}^C P(c | \mathcal{N}_k(x)) \log_2 P(c | \mathcal{N}_k(x)) \quad (2)$$

Here, we define $0 \log_2 0 := 0$. When $h(x) \rightarrow 0$, it implies that the k neighbors almost exclusively belong to the same class; the point resides in a “pure” region far from the boundary. When $h(x) \rightarrow \log_2 k$, it indicates that the neighbors originate from k (or more) completely different classes, placing the point on a highly chaotic and ambiguous decision boundary.

3.1.3. From Local to Global Boundary Entropy (GBE)

The expectation of the local boundary entropy across all samples in a dataset constitutes the Global Boundary Entropy (GBE), which measures

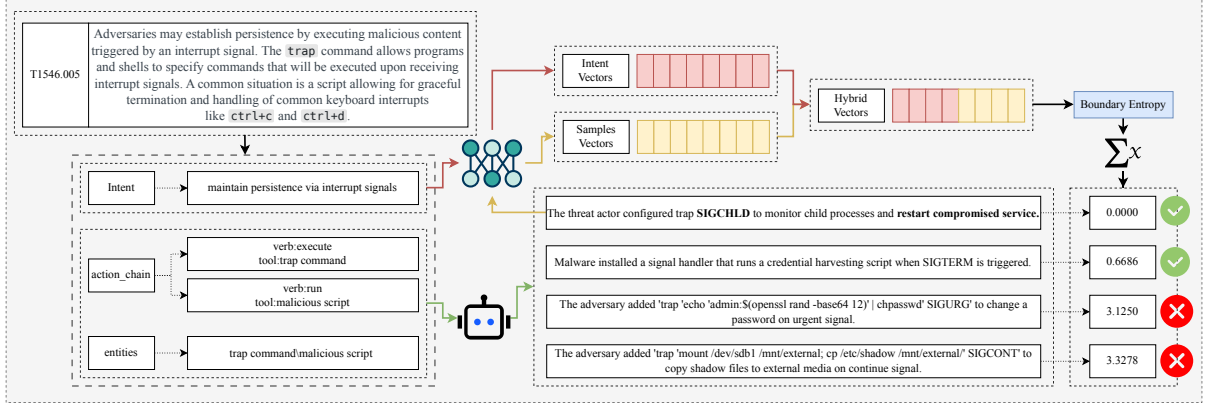


Figure 1: Overview of the Proposed Methodology

the average boundary ambiguity of the entire dataset.

Global Boundary Entropy (GBE): Given a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, its Global Boundary Entropy $H_{bd}(\mathcal{D})$ is the empirical mean of all local boundary entropies:

$$H_{bd}(\mathcal{D}) := \mathbb{E}_{(x,y) \sim \mathcal{D}}[h(x)] \approx \frac{1}{N} \sum_{i=1}^N h(x_i) \quad (3)$$

Our core hypothesis is that for a high-quality dataset, H_{bd} should be minimized.

3.1.4. Analytical Motivation: Connecting GBE with Generalization Error

To provide an analytical motivation for the minimization of H_{bd} , we establish an intuitive link to the classifier’s ultimate goal: bounding the Generalization Error (E_f).

The generalization error E_f refers to the expected error rate of model f on unseen real data. We outline this intuition-supported bound as follows:

Step 1: Defining the “Fuzzy Boundary Region”. We define a region B_ϵ containing all points with local entropy higher than a threshold ϵ :

$$B_\epsilon := \{x \in \mathcal{X} : h(x) \geq \epsilon\} \quad (4)$$

This region represents all “hard-to-classify” points in the data space.

Step 2: Connecting GBE to the Size of B_ϵ . The “size” of a region is statistically measured by its probability mass $P(B_\epsilon)$. According to Markov’s Inequality, for any non-negative random variable X and constant $a > 0$, $P(X \geq a) \leq \mathbb{E}[X]/a$. In our

context, h_x is the variable and $a = \epsilon$. Thus:

$$P(B_\epsilon) = P(h(x) \geq \epsilon) \leq \frac{\mathbb{E}[h(x)]}{\epsilon} \quad (5)$$

Substituting the definition of GBE, we obtain the key inequality:

$$P(B_\epsilon) \leq \frac{H_{bd}}{\epsilon} \quad (6)$$

This implies that the upper bound of the fuzzy boundary region’s size is directly controlled by H_{bd} . Reducing H_{bd} effectively compresses the ambiguous boundary.

Step 3: Connecting B_ϵ to Generalization Error. In statistical learning theory, the generalization error is bounded by the empirical error $E_N(f)$ plus a complexity term. Refined Margin Theory suggests this bound is closely related to the probability mass of low-margin samples ($P(\text{margin} \leq \rho)$). Our $P(B_\epsilon)$ serves as a data-driven proxy for this low-margin mass. A high-entropy region almost always corresponds to a low-margin region. Thus, we can construct an intuition-supported generalization bound as:

$$E_f \leq E_N(f) + \text{Complexity Term} + \gamma \cdot P(B_\epsilon) \quad (7)$$

where γ is a loss-related constant. Combining the inequalities from Step 2 and Step 3 yields:

$$E_f \leq E_N(f) + \text{Complexity Term} + \gamma \cdot \frac{H_{bd}}{\epsilon} \quad (8)$$

This derivation suggests that Global Boundary Entropy H_{bd} can be viewed as an additive penalty term representing boundary complexity within the upper bound of the generalization error.

3.1.5. Conclusion

The above derivation provides a reasoned analytical motivation for our methodology: minimizing Global Boundary Entropy (H_{bd}) serves as a strong heuristic to potentially reduce the generalization error (E_f). Consequently, the proposed BEDR framework—which actively screens and generates low-local-entropy samples to minimize H_{bd} —is an intuition-supported and theoretically guided approach to optimizing dataset quality.

3.2. Hybrid Vector Representation

3.2.1. Structured Semantic Disentanglement & Knowledge-Enhanced Prompting Based on Action-Chain Extraction

Traditional TTP data augmentation methods often directly input the raw technical description (Description) as context into Large Language Models (LLMs) Tao et al. (2022). However, this unstructured input suffers from severe “Instruction Ambiguity.” Technical descriptions typically contain multiple attack variants, defense bypass methods, and complex contextual conditions. Consequently, LLMs struggle to autonomously determine the focus of generation, which easily leads to “Mode Collapse” (i.e., repeatedly generating the single most common scenario found in the description) or “Hallucination” Zhao et al. (2024) (generating noise samples that deviate from the core intent).

To resolve this challenge, we propose a Structured Semantic Disentanglement mechanism. Instead of using raw text directly, we pre-utilize an LLM as a “knowledge parser” to map the unstructured description D_c into a structured attack graph $G_c = \{I, A, E\}$. The core value of this process lies in its role as the foundation for “Structured Knowledge-Enhanced Prompting,” which specifically comprises two key dimensions:

1) Intent Anchoring (I): We extract the high-level semantics describing the attacker’s final tactical goal (e.g., “maintain persistence via interrupt signals”). In subsequent generation, this is used as a Global Constraint, forcing the LLM to ensure all generated samples serve this core intent, thereby preventing Semantic Drift.

2) Action-Chain Guided Diversity (A): We parse the complex attack process into a series of discrete “Verb-Tool Pairs,” formalized as $A = \{(v_1, t_1), (v_2, t_2), \dots, (v_n, t_n)\}$. To achieve Controllability, in prompt engineering, we treat each specific (v_i, t_i) pair as an independent “Generation Seed.”

For example, for technique T1548, rather than generally asking the LLM to “generate samples,” we instruct it individually: “generate a sample using v_1 :modify against t_1 :sudoers” or “generate a sample using v_2 :bypass against t_2 :UAC.” This fine-grained control forces the LLM to traverse every potential attack path in the technical description, rather than remaining at the superficial level, thereby achieving target Coverage.

Through this mechanism, we transform the LLM generation process from “black-box groping” to “white-box gap-filling.” The structured graph G_c serves not only as a Metric Benchmark for subsequent hybrid vector calculations but also as a Navigation Map during the generation phase, ensuring that the generated samples maintain strict semantic Controllability while possessing high Diversity.

To intuitively illustrate how we guide the LLM using the constructed knowledge graph, the specific prompt template employed in our framework is presented in Figure 2.

Role Definition: As a cybersecurity expert, your task is to generate [N] diverse, realistic, and concise example sentences for the ATT&CK technique [Tech_ID].

Constraint 1: Key Objective (Intent Anchoring I)
“[Official_Intent_Text]”

Constraint 2: Action-Chain Guidance (A)
Generate sentences that clearly describe one of these core actions:

- [Verb_1]: [Tool_1]
- [Verb_2]: [Tool_2]
- ...

CRITICAL INSTRUCTIONS:

1. FOCUS: Each sentence must be a clear example of ONE specific action, not a long story or a combination of many actions.
2. FORMAT: Output ONLY a valid JSON list of strings.

Figure 2: The structured knowledge-enhanced prompt template used in BEDR. The template dynamically injects the official intent (I) as a global constraint and utilizes the extracted action chain (A) to guide the diversity of generated samples.

3.2.2. Hybrid Vector Representation

In Section 3.1, we demonstrated that minimizing Global Boundary Entropy (H_{bd}) is an effective strategy for improving generalization. However, to

reliably calculate H_{bd} , a feature space with high discrimination capability is a prerequisite.

Our preliminary analysis indicates that standard Plain Sentence Embeddings ($s_i = \text{Embed}(s_i)$) exhibit deficiencies in TTP tasks. Specifically, due to the substantial shared general semantics between parent-child techniques and similar procedures, the plain vector space $\mathcal{S}$ presents a high degree of Semantic Aliasing. In this relatively chaotic space, samples originally belonging to different categories lie extremely close to each other, leading to generally inflated and unreliable calculation results for Local Boundary Entropy $h(x)$. Directly filtering in this space would likely result in the erroneous rejection of valid samples. To address this, we propose constructing a Hybrid Vector Scaffold ($\mathcal{Z}$). This scaffold is a temporary, internal feature space designed to provide a clearer and more robust metric frame of reference for accurate entropy calculation.

The core philosophy of the hybrid vector $\mathbf{z}_i$ is to combine the Micro-level Uniqueness of the sample (raw sentence text) with the Macro-level Stability of the category (official intent). We extract the intent anchor I from the official description text to serve as a category guidance anchor. For a sample s_i belonging to category c , we calculate its hybrid vector $\mathbf{z}_i$ as follows:

Let $v_{I_c} = \text{Embed}(I_c)$ be the embedding of the official intent for class c , and $v_{s_i} = \text{Embed}(s_i)$ be the embedding of the raw sentence. Then $\mathbf{z}_i$ is defined as their L2-normalized sum:

$$\mathbf{z}_i := \frac{v_{I_c} + v_{s_i}}{\|v_{I_c} + v_{s_i}\|_2} \quad (9)$$

This combination leverages the strong classification information of I_c to artificially pull all sample vectors v_{s_i} belonging to the same class c towards a common intent center v_{I_c} . This effectively enhances Intra-class Cohesion and significantly improves Inter-class Separation in the feature space, thereby creating a metric space $\mathcal{Z}_{scaffold}$ that is far more suitable for boundary analysis.

We apply the above formula to all samples in the original dataset $\mathcal{D}_{orig}$ to pre-calculate their hybrid vectors, forming the reference space $\mathcal{Z}_{ref}$:

$$\mathcal{Z}_{ref} = \{(\mathbf{z}_i, y_i)\}_{i=1}^{N_{orig}} \quad (10)$$

This space serves as the sole frame of reference for all our subsequent boundary entropy calculations. When the LLM generates a new candidate

sample s_{cand} , we calculate its hybrid vector $\mathbf{z}_{cand}$ using the same formula and then project it into $\mathcal{Z}_{ref}$ to calculate its local boundary entropy $h(\mathbf{z}_{cand})$.

It is important to emphasize that this hybrid vector $\mathbf{z}_i$ is an internal tool of the BEDR framework, restricted solely to providing accurate boundary entropy $h(\mathbf{z}_i)$ for sample screening. It is not used for the final classification model training or evaluation, thereby ensuring the universality and fairness of the final assessment results.

3.3. Multi-Pass Adaptive Oversampling

Using a fixed strict threshold may reject valuable but naturally ambiguous samples, while a loose threshold introduces noise. We design a Multi-Pass Real-time Adaptive Oversampling algorithm, the detailed procedure of which is outlined in Algorithm 1.

The process consists of four passes (Pass = 1 to 4). In each pass, we set a target sample count τ_{pass} and a maximum relaxation threshold ϵ_{max} (e.g., $\tau_{pass} \in \{50, 10, 5\}$, $\epsilon_{max} \in \{3.0, 3.5, 3.8\}$). Before each generation attempt $t > 1$, the system dynamically adjusts the acceptance standard ϵ_c based on the quality of rejected samples:

1) Initial Strict Threshold: For each class c , the threshold is initialized to a strict value $\epsilon_c \leftarrow \epsilon_{strict}^{(p)}$ (e.g., 1.0 for the first pass, relaxed to 2.3 for the final pass), ensuring priority for golden samples in early stages.

2) Real-time Assessment: During the generation loop, we calculate the cumulative success rate $r_{success} = N_{acc}/N_{cand}$.

3) Intelligent Adjustment: If $r_{success}$ falls below a trigger value τ_{rate} (e.g., 0.20) and sufficient rejected samples exist ($\mathcal{H}_{rejected} \neq \emptyset$), we calculate a ‘‘suggested threshold’’ $\epsilon_{suggest}$. Unlike static scaling, we employ an incremental update strategy relative to the current threshold ϵ_c :

$$\epsilon_{suggest} = \epsilon_c + 0.5 \times (h_{med_rej} - \epsilon_c) \quad (11)$$

where $h_{med_rej} = \text{median}(\mathcal{H}_{rejected})$. This formula moves the threshold halfway towards the median entropy of the rejected samples.

4) Non-decreasing Principle: The final threshold is updated as:

$$\epsilon_c \leftarrow \max(\epsilon_c, \min(\epsilon_{max}, \epsilon_{suggest})) \quad (12)$$

This rule ensures the threshold strictly maintains or increases its value (‘‘climbing’’), preventing regression while capped by ϵ_{max} , effectively balancing

Algorithm 1 Multi-Pass Adaptive Oversampling with Boundary Entropy

Require: Dataset $\mathcal{D}$, Pass Configurations
 $\{\tau_p, \epsilon_{strict}^{(p)}, \epsilon_{max}^{(p)}\}_{p=1}^4, \tau_{rate} = 0.2$
 Ensure: Optimized Dataset $\mathcal{D}_{opt}$

```

1:  $\mathcal{D}_{opt} \leftarrow \mathcal{D}$ 
2: for pass  $p = 1$  to 4 do
3:    $\mathcal{C}_{target} \leftarrow \{c \in \mathcal{C} \mid |\mathcal{D}_c| < \tau_p\}$ 
4:   for each class  $c \in \mathcal{C}_{target}$  do
5:      $\epsilon_c \leftarrow \epsilon_{strict}^{(p)}; \mathcal{H}_{rej} \leftarrow \emptyset$ 
6:     while  $|\mathcal{D}_c| < \tau_p$  and not max_retries do
7:       Step 1: Real-time Adaptation
8:        $r_{succ} \leftarrow N_{acc}/N_{total\_generated}$ 
9:       if  $r_{succ} < \tau_{rate}$  and  $\mathcal{H}_{rej} \neq \emptyset$  then
10:         $\epsilon_{sugg} \leftarrow \epsilon_c + 0.5 \times (\text{Median}(\mathcal{H}_{rej}) - \epsilon_c)$ 
11:         $\epsilon_c \leftarrow \max(\epsilon_c, \min(\epsilon_{max}^{(p)}, \epsilon_{sugg}))$ 
12:      end if
13:      Step 2: Structured Generation
14:       $\mathcal{S}_{cand} \leftarrow \text{LLM}(\text{Prompt with } G_c)$ 
15:      Step 3: Boundary Entropy Filtering
16:      for  $s \in \mathcal{S}_{cand}$  do
17:        Compute Hybrid Vector  $\mathbf{z}_s$  using Eq. (10)
18:        Calculate Entropy  $h(\mathbf{z}_s)$  in Reference Space  $\mathcal{Z}_{ref}$ 
19:        if  $h(\mathbf{z}_s) < \epsilon_c$  then
20:           $\mathcal{D}_{opt} \leftarrow \mathcal{D}_{opt} \cup \{s\}; N_{acc} \leftarrow N_{acc} + 1$ 
21:        else
22:           $\mathcal{H}_{rej} \leftarrow \mathcal{H}_{rej} \cup \{h(\mathbf{z}_s)\}$ 
23:        end if
24:      end for
25:    end while
26:  end for
27: end for
28: return  $\mathcal{D}_{opt}$ 
```

the trade-off between sample diversity and boundary clarity for stubborn classes.

3.4. Entropy-Based Selective Undersampling

In dealing with long-tail distributions, addressing the redundancy and noise in head categories (Majority Classes) is just as critical as supplementing tail categories (Minority Classes). Traditional Random Undersampling, while effective for balancing sample quantities, is prone to inadvertently discarding representative core samples or retaining ambiguous samples located at decision boundaries (i.e., high-entropy samples), thereby compromising the structural integrity of the feature space.

As the undersampling component of the BEDR framework, we designed a Boundary-Aware Cleansing Strategy. Echoing the logic of “generating low-entropy samples” in the oversampling phase, the

primary objective of this phase is to “eliminate high-entropy samples.”

The specific algorithmic procedure is as follows:

1) Majority Identification and Threshold Setting: First, we define a majority class retention threshold τ_{maj} (set to 80 in this study based on empirical analysis). Any category c with a sample count $|\mathcal{D}_c| > \tau_{maj}$ is marked as a majority class requiring processing.

2) Hybrid-Space Entropy Ranking: To accurately assess the “retention value” of each sample, we explicitly leverage the Hybrid Vector Scaffold constructed in Section 3.2. For every existing sample x_i in a majority class c , we calculate its local boundary entropy $h(\mathbf{z}_i)$ within this hybrid reference space.

- Low Entropy ($h(\mathbf{z}_i) \rightarrow 0$): Implies that the sample is tightly surrounded by instances of the same class. It resides in the “core zone” of the category, serving as a Prototype that represents the category’s distinct features, and thus should be retained.
- High Entropy ($h(\mathbf{z}_i) \rightarrow \text{High}$): Implies that the sample is located in inter-class overlap regions or acts as an outlier. It represents potential Noise that causes classification confusion, and thus should be prioritized for removal.

3) Top-K Truncation: We sort all samples in category c in ascending order based on their boundary entropy $h(\mathbf{z}_i)$. Subsequently, we retain only the top τ_{maj} samples with the lowest entropy values, discarding the remaining tail high-entropy samples:

$$\mathcal{D}_c^{new} = \{x_i \in \mathcal{D}_c \mid \text{rank}(h(\mathbf{z}_i)) \leq \tau_{maj}\} \quad (13)$$

Through this “Top-K Truncation” undersampling strategy (a process of extracting the essence and discarding the dross), the BEDR framework not only reduces the dominance of head categories in terms of quantity but also achieves the Sharpening of the feature space in terms of quality. By systematically removing ambiguous boundary samples, it artificially widens the inter-class Margin, thereby further minimizing the Global Boundary Entropy (H_{bd}) of the dataset.

4. Experiments

To comprehensively validate the effectiveness of the BEDR framework in addressing the challenges of “long-tail distribution” and “boundary ambiguity” in fine-grained Cyber Threat Intelligence (CTI)

mapping, this section conducts an in-depth analysis centering on the following five core Research Questions (RQs):

- RQ1 (Dataset Quality): Does the dataset constructed via the BEDR strategy exhibit superior balance and distributional characteristics compared to the naive augmentation strategy? (Note: Naive augmentation refers to directly utilizing prompt engineering to guide LLMs in generating samples, as seen in [33]).
- RQ2 (Core Effectiveness): Compared to the baseline (no augmentation) and naive LLM augmentation, can BEDR significantly enhance classification performance, particularly for zero-shot and few-shot categories?
- RQ3 (Mechanism Validation): Under the premise of controlling for identical sample size, is BEDR’s “boundary entropy filtering” mechanism more effective than mere “quantity accumulation”? (i.e., the verification of “Quality vs. Quantity”).
- RQ4 (Component Contribution): What are the individual contributions of the framework’s various components (sampling strategy, metric space, prompt engineering) to the final performance? (Ablation Study).
- RQ5 (Comparative Superiority): Compared with existing traditional resampling techniques (e.g., SMOTE-based methods) and text augmentation techniques (e.g., EDA-based methods), does BEDR possess significant advantages?

4.1. Dataset Construction

To comprehensively verify the effectiveness of the proposed BEDR framework in mitigating the problems of class imbalance and boundary ambiguity, this study constructed a multi-dimensional dataset system. The ultimate goal is to establish a comprehensive Atomic Action Benchmark Dataset for fine-grained threat intelligence mapping, covering a baseline, naive LLM generation, SMOTE augmentation, EDA augmentation, and our proposed BEDR optimization method.

4.1.1. Original Corpus and Baseline Dataset ($D_{baseline}$)

This paper uses the MITRE ATT&CK v17.1 official knowledge base as the sole corpus source, with

mapping unstructured text to 679 technical categories (containing 211 parent techniques and 468 sub-techniques) as the classification target.

Only the “procedure examples” of each technical category in ATT&CK are used as basic training samples; the “description” field serves only as background knowledge for extracting official intents and guiding LLM generation, and is not directly used as a training sample. After deduplication and cleaning, the baseline dataset $D_{baseline}$ was constituted. As shown in Table 2 and Figure 3, this dataset presents extreme long-tail distribution characteristics: as many as 105 categories (accounting for about 15.5%) are zero-shot categories. The median sample size is extremely small, while the maximum category sample size is heavily skewed, and the sample distribution is extremely imbalanced. This distribution characteristic is the fundamental reason for the failure of traditional models on small-sample categories.

4.1.2. Construction of Control and Experimental Group Datasets

In order to fairly compare the effectiveness of different augmentation strategies under the same scale, we constructed the following augmented training datasets.

1) Naive LLM Resampling Training Dataset (D_{train_Naive}): Utilizing LLM (DeepSeek-Chat) to generate samples directly based on the technical description “description”. For small-sample categories, generation continued until the target quantity was supplemented (oversampling); for large-sample categories, a Random Undersampling strategy was adopted to reduce to the target threshold. This process did not perform any quality filtering based on boundary entropy. The obtained D_{train_Naive} has a total sample volume of 33,804. Although it eliminated zero-shot categories and achieved a high degree of balance in quantity, its sample generation did not undergo quality screening, and the distribution presents an artificial feature of “forced uniformity.”

2) BEDR Resampling Training Dataset (D_{train_BEDR}): Adopting the Boundary-Entropy-driven Resampling framework proposed in this paper to construct the high-quality atomic action benchmark. In the oversampling process, the “Hybrid Vector + Boundary Entropy” mechanism was utilized for four rounds of adaptive generation and filtering. In the undersampling process, large categories were cleansed based on boundary

entropy. Finally, D_{train_BEDR} was obtained, with a total sample volume of 17,162. Compared to the forced filling of the naive strategy, the distribution of BEDR aligns better with the quality-first principle of “strict quality control over quantity,” and while improving imbalance, it maximizes the maintenance of the purity of the feature space.

4.1.3. In-the-wild Independent Test Set (D_{test})

To address potential data leakage and the zero-shot evaluation paradox, it is crucial that the test set is completely independent of the augmentation pipeline. We constructed a strictly held-out, in-the-wild test set D_{test} comprising 4,291 samples.

These samples were carefully curated from external, real-world CTI sources, including security vendor blogs (e.g., Mandiant, Palo Alto Unit 42), public incident reports, and CVE vulnerability descriptions, covering the 679 technical categories. Importantly, the sample distribution of D_{test} naturally follows the inherent long-tail characteristics of real-world threats. Rare or zero-shot techniques in the official ATT&CK examples also appear with lower frequency in this in-the-wild set.

By ensuring that D_{test} contains no synthetic samples generated by the LLM during the training phase, we guarantee that all models are evaluated fairly. Furthermore, to eliminate any potential data leakage from the original corpus, we performed strict semantic deduplication. Any procedural examples in the official ATT&CK baseline that exhibited high semantic overlap with our collected in-the-wild test samples were rigorously removed. This purification process resulted in the final baseline training set $D_{train_Baseline}$ comprising exactly 12,800 strictly independent samples. Consequently, the performance gains observed on the 105 zero-shot categories strictly reflect the model’s authentic capability to generalize and recognize previously unseen, real-world attack behaviors. The specific configuration of the data splits is detailed in Table 3.

4.2. Experimental Design

4.2.1. Model Architecture and Core Variants

To answer RQ1 through RQ3, we designed hierarchical comparative model groups. All models are based on the same Multi-Layer Perceptron (MLP) architecture, consisting of an input layer (768 dimensions), a single hidden layer (256 dimensions with ReLU activation), and an output layer (679 dimensions). The training process uniformly employs the Adam optimizer (learning rate $2e-5$) for

800 epochs. To strictly ensure the statistical robustness of our findings and mitigate the influence of random initialization, all models were trained and evaluated 5 times using different random seeds (ranging from 42 to 46). The final performance metrics are reported as the mean $\pm$ standard deviation across these 5 independent runs.

It is worth noting that this study deliberately selected a structurally simple MLP network, rather than fine-tuning complex pre-trained models (such as BERT), as the validation classifier. This is based on two core considerations:

- **Isolation of Variables:** The focus of this study is to verify the quality of the dataset constructed by the BEDR framework (i.e., the separability and purity of features). Complex deep models often possess strong fitting capabilities, which might mask noise issues in the data through overfitting. Using a standardized lightweight model allows it to act as an “honest probe,” most sensitively quantifying the pure performance gains brought by different data augmentation strategies.
- **Baseline Fairness:** This aims to demonstrate the universality of the BEDR framework, proving that performance improvements stem from improved data quality rather than the dividends of a specific model architecture.

Regarding feature representation, all models uniformly use a general pre-trained text embedding model (text-embedding-v4) to generate standard Plain Sentence Embeddings as input during the evaluation phase, eliminating interference from differences in feature engineering.

We constructed the following four groups of core models to verify the effectiveness of the BEDR framework:

- **Model B (Baseline):** Trained using only the original dataset $D_{train_Baseline}$. This model represents the performance lower bound of existing imbalanced data without any augmentation, used to assess the severity of the “cold start” problem.
- **Model C (Naive LLM - Full):** Trained using the full naive augmented dataset D_{train_Naive} . This model represents the “Quantity-First” strategy, aiming to verify the performance ceiling brought by brutally expanding sample size via LLM alone.

Table 2: Comparison of Statistical Indicators for Experimental Training Datasets

Dataset	Total Samples	Zero-Shot Classes	Min Samples	Max Samples	Mean	Median	Std Dev
$D_{train_Baseline}$	12,800	105	0	480	18.85	4.0	48.20
D_{train_Naive}	33,804	0	47	80	49.78	50.0	7.85
D_{train_BEDR}	17,162	0	5	80	25.28	21.0	18.50

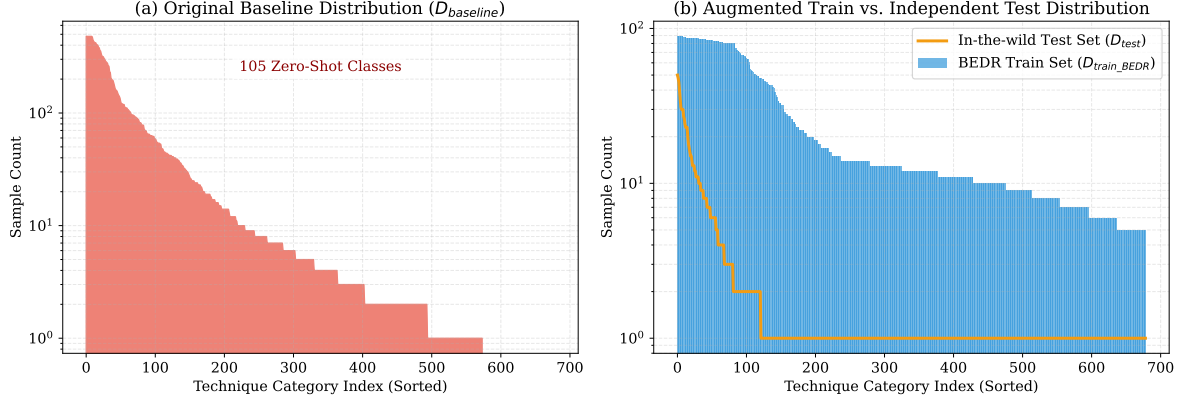


Figure 3: Consistency Analysis of Sample Distribution. (a) illustrates the severe long-tail distribution of the baseline dataset; (b) compares the distributions of the BEDR augmented training set and the strictly independent in-the-wild test set, demonstrating that the evaluation reflects real-world threat frequencies.

- Model C' (Naive LLM - Subset): This is a critical control group. We performed stratified sampling from D_{train_Naive} to construct a subset with a size nearly identical to the BEDR training set ($|D_{C'}| \approx |D_A|$). This model is used to eliminate the confounding variable of increased sample quantity, thereby precisely quantifying the difference in sample quality brought by the boundary entropy filtering mechanism.
- Model A (BEDR Optimized): Trained using the BEDR optimized dataset D_{train_BEDR} proposed in this paper, representing the "Quality-First" strategy.
- V1 (+OVR only): Oversamples only small-sample categories (target is 50, using dynamic adaptive method), without processing large categories.
- V2 (+UVR only): Undersamples only large-sample categories (reduced to 80), without supplementing small categories.
- V3 (+Random Balance): Combines oversampling and undersampling but does not use boundary entropy filtering (i.e., randomly accepts generated samples and randomly removes original samples). This represents the traditional "Quantitative Balance" benchmark.

4.2.2. Ablation Study Design

To comprehensively deconstruct the performance sources of the BEDR framework and verify the independent contributions of core components (sampling strategy, metric space, prompt engineering), we designed ablation experiments in two dimensions.

Part 1: Impact of Sampling Strategies. First, we verify the fundamental roles of oversampling and undersampling in addressing imbalance. We constructed three basic variants:

Part 2: Impact of Core Quality Mechanisms. After verifying the necessity of sampling strategies, we further explore the effectiveness of BEDR's two core innovations—the "Hybrid Vector Scaffold" and "Structured Semantic Disentanglement." To this end, we introduced two specific control variants based on V3:

- V4 (w/o Hybrid Scaffold): A "degenerated" BEDR. It retains the structured prompt generation and boundary entropy filtering processes, but when calculating boundary entropy, it

Table 3: Configuration of Independent Test Set and Comparative Training Sets

Dataset	Sample Size	Target Use	Source / Description
D_{test}	4,291	Unified Evaluation	External (Blogs, Reports, CVEs) - In-the-wild data
D_{train_BEDR}	17,162	Model A Training	BEDR Augmented Atomic Action Benchmark
D_{train_Naive}	33,804	Model C Training	Naive LLM Augmented Training Pool
$D_{train_Baseline}$	12,800	Model B Training	Original ATT&CK Procedural Examples (Deduplicated)

does not use hybrid vectors and instead directly uses plain sentence embeddings. This verifies whether the hybrid vector space can measure sample quality more accurately than the ordinary vector space.

- V5 (w/o Structured Prompt): Another “de-generated” BEDR. It retains hybrid vectors and boundary entropy filtering, but does not use Action-Chain guidance during the generation phase, instead using naive prompting. This verifies whether “Structured Prompting” generates original material with greater diversity and separability than “Naive Prompting.”

4.2.3. External Baselines and Feasibility Protocol

To answer RQ5, we performed a horizontal comparison between Model A and representative advanced augmentation methods in the field. The comparison baselines include:

- Text Augmentation: EDA Wei and Zou (2019), representing traditional rule-based NLP augmentation strategies.
- Vector Oversampling: SMOTE Chawla et al. (2002), representing classic feature space linear interpolation strategies.
- Hybrid Resampling: SMOTE-IPF Sáez et al. (2015) and SMOTE-RkNN Zhang et al. (2022), representing advanced resampling strategies combining generation and cleaning mechanisms.

Construction of Feasible Subset: It is worth noting that the aforementioned traditional methods face a fundamental failure scenario in threat intelligence mapping tasks: Zero-Shot categories. Our baseline dataset has 105 categories with zero samples. SMOTE and its variants rely on k-Nearest Neighbors (k-NN) interpolation in the feature space and cannot generate new samples without any seed samples; EDA relies on perturbations of existing

text and similarly cannot handle empty sets. Furthermore, SMOTE-like algorithms mathematically require each category to have at least 2 samples to define a geometric neighborhood. Although the BEDR framework can naturally handle zero-shot and one-shot scenarios through the prior knowledge of LLMs, to ensure strict fairness in the comparison for RQ5, we constructed a specific “Non-Singleton Subset” (denoted as $D_{\geq 2}$), excluding categories with original sample counts $N < 2$. All external comparison experiments in this section are conducted on this feasible subset. This protocol effectively isolates the evaluation of “generation quality” from “cold start capability,” ensuring the rigor of the comparative conclusions. The statistics are shown in Table 4.

4.2.4. Evaluation Metrics

Given that the TTP dataset in this study exhibits extreme long-tail distribution characteristics, traditional Accuracy metrics can be easily dominated by head categories with large samples, thereby masking the model’s failure on tail categories. To evaluate model performance comprehensively and objectively from different dimensions, this study adopts the following four core metrics:

- Precision & Recall: In CTI analysis, these have specific practical significance. Precision reflects the reliability of the model’s predictions (i.e., “how many of the alerts issued by the model are real TTPs”), where high precision helps reduce operational fatigue from false alarms; Recall reflects the coverage capability (i.e., “how many real TTPs are successfully identified by the model”), where high recall is crucial for avoiding the fatal security risk of missing detections. We calculate the average precision and average recall across all categories.
- Micro-F1 Score: This metric calculates the global F1 value by accumulating True Positives (TP), False Positives (FP), and False Negatives

Table 4: Statistics of Feasible Subset Datasets for External Experiments

Dataset	Total Samples	Class Count	Description
$D_{train_base_feasible}$	12,734	487	Baseline for RQ5 ($D_{\geq 2}$)
$D_{train_test_feasible}$	3,582	487	Test Set for RQ5
$D_{train_EDA_feasible}$	29,541	487	Oversampling Comparison: EDA
$D_{train_SMOTE_feasible}$	28,796	487	Oversampling Comparison: SMOTE
$D_{train_SMOTE-IPF_feasible}$	21,356	487	Resampling Comparison: SMOTE-IPF
$D_{train_SMOTE-RkNN_feasible}$	22,431	487	Resampling Comparison: SMOTE-RkNN
$D_{train_BEDR_feasible}$	14,322	487	BEDR Adjusted Training Set for RQ5

(FN) across all categories. Since it is calculated based on the total number of samples, Micro-F1 effectively measures the model’s classification performance on the overall data distribution, reflecting the model’s fit to large-sample categories.

- **Macro-F1 Score:** This metric first calculates the F1 score for each category separately and then computes the arithmetic mean. Since Macro-F1 treats every category equally (regardless of sample size), it is not “diluted” by large-sample categories. Therefore, in long-tail distribution scenarios, Macro-F1 is the core metric that most sensitively reflects the model’s generalization ability on “Zero-Shot” and “Few-Shot” categories, and serves as the key basis for verifying whether the BEDR framework proposed in this study effectively solves the imbalance problem.

4.3. Results and Analysis

4.3.1. RQ1-RQ3: Core Effectiveness (Part 1 Metric Evaluation)

To answer RQ1 (Dataset Quality), RQ2 (Core Effectiveness), and RQ3 (Quality vs. Quantity), we conducted a comprehensive performance evaluation of the baseline model (Baseline), naive augmentation model (Naive), and BEDR optimized model on the unified test set (D_{test}). The experimental results are presented in Table 5.

From the data in the table, the following key conclusions can be drawn:

1) **Breaking the Cold Start Bottleneck:** BEDR Significantly Outperforms Baseline (Answering RQ2). Compared to Model B which uses only original data, Model A (BEDR) achieved substantial breakthroughs across all evaluation metrics. BEDR increased Macro-F1 from 26.78% to 32.76%, an absolute improvement of 5.98% and a relative

improvement of 22.3%. Although a Macro-F1 of 32.76% might appear numerically modest, it must be interpreted in the context of the task’s extreme difficulty: this is a fine-grained classification task targeting 679 categories, containing highly semantically overlapping parent-child techniques. For reference, the accuracy of random guessing is only about 0.15%. The 22.3% relative improvement achieved by BEDR over the baseline represents a significant breakthrough in such a high-dimensional classification space, proving its core effectiveness in solving “Zero-Shot” and “Long-Tail” distribution problems. It successfully transformed previously unlearnable zero-shot categories into a feature space comprehensible to the model. Micro-F1 also increased from 40.41% to 45.09%, indicating that BEDR not only optimized tail categories but also enhanced the model’s generalization ability regarding the overall sample distribution.

2) **Mechanism Validation:** The Core Value of Boundary Entropy Filtering (Answering RQ3 - Part 1). The comparison between Model A and Model C’ (Naive Subset) is crucial for validating the effectiveness of BEDR’s core mechanism. Since both maintained consistency in training sample scale (controlling the quantity variable), the performance difference directly reflects the difference in sample quality. Under the same data scale, Model A’s Macro-F1 (32.76%) was significantly higher than Model C’ (28.17%), a difference of 4.59%. This indicates that samples generated by naive LLM contain a large amount of ineffective or low-quality noise, whereas BEDR successfully screened out high-value samples located near the decision boundary through the introduction of the “Hybrid Vector Scaffold” and “Boundary Entropy Filtering” mechanisms. This “semantic purification” process is the fundamental source of BEDR’s superior performance.

3) **Quality Over Quantity:** Overcoming Mode Collapse (Answering RQ3 - Part 2). Even when

Table 5: Comparison of Core Experimental Results (Mean $\pm$ Std. Dev. over 5 runs)

Dataset	Model	Precision	Recall	Macro-F1	Micro-F1
D_{train_BEDR}	Model A (BEDR)	$33.98 \pm 0.42\%$	$34.05 \pm 0.38\%$	$32.76 \pm 0.35\%$	$45.09 \pm 0.45\%$
D_{train_Naive}	Model C (Naive Full)	$36.16 \pm 0.55\%$	$30.43 \pm 0.62\%$	$28.72 \pm 0.58\%$	$43.32 \pm 0.50\%$
D_{train_Naive}	Model C' (Naive Subset)	$32.16 \pm 0.60\%$	$28.43 \pm 0.55\%$	$28.17 \pm 0.52\%$	$41.32 \pm 0.48\%$
$D_{train_Baseline}$	Model B (Baseline)	$28.79 \pm 0.45\%$	$27.80 \pm 0.40\%$	$26.78 \pm 0.38\%$	$40.41 \pm 0.42\%$

compared to Model C (Naive Full) which has a larger sample size, the smaller-sized Model A still demonstrated superior comprehensive performance (Macro-F1: 32.76% vs. 28.72%). More importantly, by analyzing the relationship between Precision and Recall, we revealed the inherent defect of the naive generation strategy: the Mode Collapse phenomenon. Model C exhibited a distinct bias of Precision (36.16%) $\gg$ Recall (30.43%). This phenomenon implies severe homogenization in naive generated samples—LLMs tend to repeatedly generate “safe samples” with similar structures and simple semantics (see detailed analysis in 4.3.2). Although the model memorized these simple patterns (high precision), it failed to cover real attack variants due to a lack of diversity, leading to missed detections of complex samples (low recall). Conversely, Model A achieved a high balance between Precision (33.98%) and Recall (34.05%), with a Recall value significantly higher than Model C (+3.62%). This directly corroborates our hypothesis in the methodology (Section 3.2.1): through “Action-Chain Guidance” and “Multi-Pass Adaptive Generation,” BEDR successfully introduced richer semantic variants, effectively covering the potential feature space of attack techniques. In cyber threat detection scenarios, such High Recall implies a lower risk of missed detections, possessing higher practical value. In summary, the dataset constructed by BEDR (D_{train_BEDR}) achieved dual optimization of “High Information Density” and “High Feature Coverage” at the semantic level, proving its superiority as a high-quality data augmentation framework.

“Furthermore, robustness statistics across 5 independent runs demonstrate that BEDR achieves the lowest standard deviation (e.g., $\pm 0.35\%$ on Macro-F1) compared to naive generation ($\pm 0.58\%$), empirically proving that the boundary entropy filtering mechanism not only improves the absolute performance but also significantly stabilizes the learning process by rejecting semantic noise.”

4.3.2. RQ1-RQ3: Spatial Topology Analysis (Part 2)

To comprehensively investigate the impact of different augmentation strategies on the topology of the feature space, we conducted t-SNE visualization analysis on the semantically complex T1546 (Event Triggered Execution) family (containing 17 fine-grained sub-techniques, which has the highest boundary entropy among the top 5 family categories). Figure 4 clearly displays the differences in semantic space morphology constructed by the four strategies: Baseline, Naive (Full & Subset), and BEDR.

1) Defects of Naive Strategy: Mode Collapse and Boundary Confusion (Fig. 4b & c). Naive Full (Fig. 4b): Although clusters formed, they presented typical features of “Dense Clumps.” Samples of the same sub-technique overlapped highly with almost no internal structure. This intuitively reveals the Mode Collapse phenomenon—the LLM piled up quantity merely by repeating a few simple sentence structures, resulting in a lack of effective information increment in the feature space. Naive Subset (Fig. 4c): When we controlled the sample quantity to be consistent with BEDR, the performance of naive data deteriorated sharply. Inter-class distances became blurred (Indistinct Boundaries), and samples of different sub-techniques appeared in a “Confused” state. This indicates that naive generated samples lack sufficient semantic discriminative power; once the quantity advantage is lost, the decision boundaries they construct disintegrate rapidly.

2) Advantages of BEDR: Sparse Structure and Semantic Separation (Fig. 4d). In contrast, BEDR (Fig. 4d) exhibited an ideal morphology of “Structured Diversity,” manifested in two significant features:

- Intra-cluster Sparsity: Unlike the dense clumps of Naive, the interior of each sub-technique cluster in BEDR presented a sparse distribution. This proves high semantic heterogeneity of the generated samples—BEDR successfully

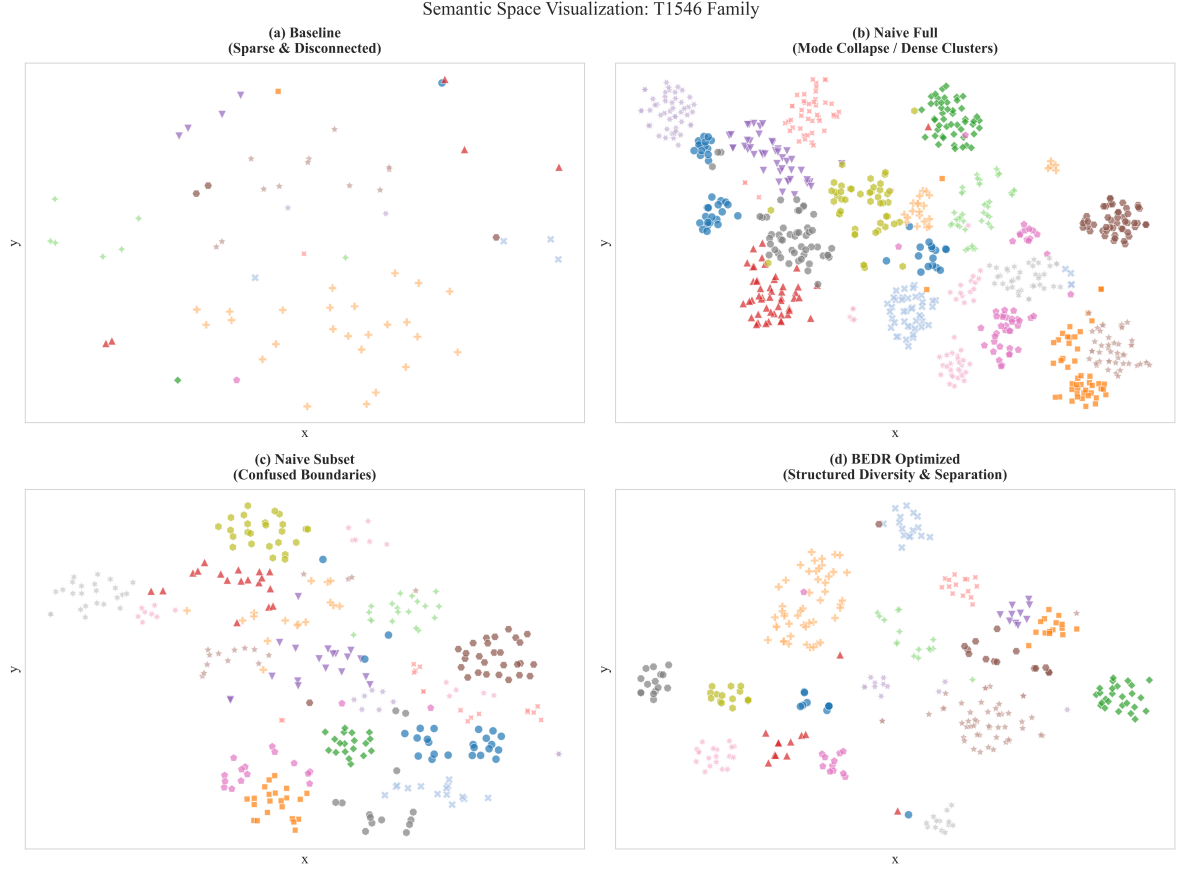


Figure 4: Visualization of Feature Space Topology for the T1546 Family. (a) Baseline shows sparsity; (b) Naive Full exhibits mode collapse (dense clumps); (c) Naive Subset shows confused boundaries; (d) BEDR demonstrates structured diversity with clear inter-class separation.

guided the model to generate diverse attack scenarios and expressions rather than simple synonym repetitions, effectively avoiding mode collapse.

- **Significant Inter-cluster Separation:** This is the most unique advantage of BEDR. Clusters of different sub-techniques maintained significant distances, with clearly visible boundaries. This is attributed to the “Structured Semantic Disentanglement” mechanism in the BEDR framework (see Section 3.2.1): by explicitly anchoring the attack intent and action chain, the algorithm artificially “pushed apart” the distances of easily confused categories in the feature space.

4.3.3. RQ1-RQ3: Case Study (Part 3)

To delve into how BEDR optimizes decision boundaries, we conducted a qualitative analysis at

the micro-level on T1546.005 (Event Triggered Execution: Trap). According to the definition of MITRE ATT&CK, the key to this technique lies in “adversaries establishing persistence by executing malicious content triggered by interrupt signals.” Table 6 shows a comparison between high-quality samples retained by BEDR and samples generated by the naive strategy but rejected due to high boundary entropy (H).

BEDR retained samples (top rows) exhibited extremely low entropy values ($H \approx 0.0 - 0.67$). These samples closely adhered to the core intent of “using signals to maintain persistence.” For instance, using SIGCHLD to restart services or SIGHUP to reload configurations perfectly fits the definition. They tightly surrounded the semantic center of the technique in the feature space. In contrast, samples generated by the naive strategy but rejected (bottom rows), although syntactically correct in us-

Table 6: Case Study: T1546.005 (Trap) Sample Analysis

Category	Entropy (H)	Sample Content	Analysis
BEDR Retained	0.0000	The threat actor configured trap SIGCHLD to monitor child processes and restart compromised services.	Pure Intent: Closely adheres to “Persistence,” using standard signals to maintain service operation.
BEDR Retained	0.6686	Malware installed a signal handler that runs a trap for SIGHUP to reload its configuration and maintain persistence.	Stable Intent: Explicitly mentions “maintain persistence,” aligning with the core goal.
Naive (Rejected)	3.0778	In a system script, the attacker inserted 'trap adduser -system... SIGSTKFLT' to create a hidden user...	Drift → T1136: Explicit “adduser” command masks the trap mechanism; semantics biased towards account creation.
Naive (Rejected)	3.1250	The adversary added 'trap echo... chpasswd SIGURG' to change a password on urgent signal.	Drift → T1098: “chpasswd” causes strong semantic bias towards account manipulation.
Naive (Rejected)	3.3278	The adversary added 'trap mount...; cp /etc/shadow SIGCONT' to copy shadow files...	Drift → T1005: File copying behavior dominates the vector embedding, leading to intent deviation.

ing the trap command, exhibited severe Semantic Drift. For example, the sample containing 'trap... adduser...' ($H = 3.0778$) had a payload core semantic of “create user.” In the vector space, this strong semantic force pulled it towards T1136 (Create Account). Since the actual tactical objectives (e.g., privilege escalation, collection) of these samples overwhelmed the trigger mechanism (persistence/execution), they fell into the inter-class fuzzy zone. BEDR prevented the classifier from confusing T1546.005 with other high-overlap techniques by filtering these high-entropy samples, ensuring not only sample purity but also that the generated data strictly served the predetermined intent of “Persistence.”

4.3.4. RQ4: Ablation Study

To deeply deconstruct the performance sources of the BEDR framework and quantify the independent contributions of core components, we designed an ablation experiment containing 5 variants. The results are summarized in Table 7.

Through longitudinal comparative analysis of Table 7, we drew key conclusions at three levels: 1) Necessity of Basic Sampling Strategies (Impact of Sampling Strategies): Comparison of V1, V2, V3 with the baseline verified the basic benefits of solving data imbalance. Oversampling played a dominant role (V1 vs. Baseline), confirming that filling the gap in zero-shot classes is the primary driver

for performance improvement. Undersampling had a minimal auxiliary effect (V2 vs. Baseline), indicating that reducing head categories alone cannot solve the fundamental difficulties of long-tail classification. V3 (Random Balance) showed limited improvement, proving that mere quantitative balance without quality control cannot build perfect decision boundaries.

2) Metric Value of Hybrid Vector Scaffold (Impact of Hybrid Vector Scaffold): V4 removed hybrid vectors and calculated boundary entropy directly in the plain sentence vector space. Compared to Model A, V4’s Macro-F1 dropped by 1.91%. In the plain vector space, semantically similar parent-child techniques (e.g., T1548.001 and T1548.002) were often extremely close, causing distortion in boundary entropy calculation. Hybrid vectors, by introducing Intent as an anchor, artificially widened inter-class distances, acting as a “High-Resolution Microscope” during data cleaning to ensure sample purity.

3) Generation Quality Contribution of Structured Prompting (Impact of Structured Prompting): V5 retained entropy filtering but removed Action-Chain guidance. V5’s Recall (31.50%) was significantly lower than Model A (34.05%), leading to a 1.56% drop in F1. This echoes the discussion on “Mode Collapse” in 4.3.1. Naive generated samples were severely homogenized; even after entropy filtering, the remaining samples lacked sufficient di-

Table 7: Ablation Study Statistical Analysis (Mean $\pm$ Std. Dev. over 5 runs)

Model Variant	Component Configuration	Macro-F1	Precision	Recall	Δ F1 vs. BEDR
Model B	Baseline (Raw Data)	26.78 $\pm$ 0.38%	28.79 $\pm$ 0.45%	27.80 $\pm$ 0.40%	-5.98%
V1	+ OVR only (Gen No-Filter)	27.85 $\pm$ 0.42%	30.15 $\pm$ 0.50%	30.10 $\pm$ 0.48%	-4.91%
V2	+ UVR only (Clean Head)	27.10 $\pm$ 0.40%	29.50 $\pm$ 0.46%	27.90 $\pm$ 0.42%	-5.66%
V3	+ Random Balance (Model C')	28.17 $\pm$ 0.52%	32.16 $\pm$ 0.60%	28.43 $\pm$ 0.55%	-4.59%
V4	w/o Hybrid Scaffold	30.85 $\pm$ 0.45%	32.50 $\pm$ 0.48%	31.20 $\pm$ 0.46%	-1.91%
V5	w/o Structured Prompt	31.20 $\pm$ 0.48%	33.20 $\pm$ 0.52%	31.50 $\pm$ 0.50%	-1.56%
Model A	Full BEDR	32.76 $\pm$ 0.35%	33.98 $\pm$ 0.42%	34.05 $\pm$ 0.38%	-

versity to cover complex attack variants. BEDR’s Structured Prompt forced the model to traverse various “Verb-Entity” combinations, guaranteeing information entropy and coverage from the source. The ablation study powerfully proved the systemic advantages of the BEDR framework: Structured Prompt is responsible for “Open Source” (providing diverse high-quality material), Hybrid Vector Scaffold for “Clear Vision” (providing precise metric standards), and Adaptive Resampling for “Throttling” (achieving dynamic balance). The three work synergistically and are indispensable.

4.3.5. RQ5: Comparison with External Baselines

The results of the horizontal comparison across methods on the feasible subset ($N \geq 2$) are shown in Table 8.

Through cross-method comparison, we reached the following profound analytical conclusions:

1) Semantic Manifold Outperforms Geometric Interpolation (BEDR vs. SMOTE Variants): Although SMOTE and its improved versions (IPF, RkNN) perform well on structured data, their performance in TTP text classification tasks was generally lower than BEDR (36.45%), and some metrics were even lower than simple text augmentation EDA (31.25%). SMOTE-like algorithms generate samples based on linear interpolation in the feature space. In the high-dimensional semantic space of NLP, the connection points between two text vectors often do not correspond to any meaningful natural language text (i.e., falling into meaningless regions outside the “Semantic Manifold”). These interpolated samples are filled with semantic noise, confusing the classifier’s decision boundaries. BEDR generated samples are essentially natural language, naturally located on the legal semantic manifold. This proves that in fine-grained classification tasks in the NLP field, semantic generation based on LLM is fundamentally superior to interpolation based on geometric features.

2) Necessity of Cleansing Mechanism (SMOTE vs. SMOTE-IPF/RkNN): Comparing standard SMOTE (29.40%) with its improved versions SMOTE-IPF (30.65%) and SMOTE-RkNN (30.95%), we found that introducing noise cleansing mechanisms indeed brought performance improvements (F1 increased by about 1.2% 1.5%). SMOTE-IPF and RkNN optimized the feature space by identifying and removing “unsafe samples” or “boundary noise” produced by interpolation. This collaterally supports the importance of the “Undersampling” and “Boundary Entropy Filtering” modules in BEDR—generating samples is only the first step; how to cleanse fuzzy samples located at the decision boundary is the key to improving performance. Although IPF/RkNN performed cleansing, they relied on simple Euclidean distance or k-NN neighborhoods. BEDR, utilizing the “Hybrid Vector Scaffold,” introduced explicit Intent as a metric scale. This “Semantic-Aware” cleansing mechanism is more penetrating than simple “Distance-Aware” cleansing, thus BEDR’s performance remained significantly leading (+5.5% or more).

3) Knowledge Injection Outperforms Rule Perturbation (BEDR vs. EDA): EDA achieved the best result among baseline methods (Macro-F1 31.25%), indicating that maintaining the semantic integrity of text (even just simple synonym replacement) is more effective than blind vector interpolation. However, this was still significantly lower than BEDR. EDA relies on rule perturbation, which is essentially a tiny “jitter” within the existing semantic space and cannot produce samples containing new information. BEDR utilizes the external knowledge base of LLM to generate attack variants that never appeared in the original data (e.g., specific tool parameters, alternative execution paths). This “Knowledge Injection” capability is the core driving force for BEDR to surpass traditional text augmentation and achieve high Re-

Table 8: Comparison with External Baselines on Feasible Subset ($N \geq 2$, Mean $\pm$ Std. Dev. over 5 runs)

Category	Dataset Model	/ Method Description	Precision	Recall	Macro-F1	Δ vs. BEDR
Baseline	$D_{train_base_feasible}$	Raw Data (No Augmentation)	$30.12 \pm 0.45\%$	$29.45 \pm 0.42\%$	$28.56 \pm 0.40\%$	-7.89%
Text Aug.	$D_{train_EDA_feasible}$	Text Augmentation (Synonym/Swap)	$33.45 \pm 0.55\%$	$31.80 \pm 0.50\%$	$31.25 \pm 0.48\%$	-5.20%
Vector Gen.	$D_{train_SMOTE_feasible}$	Vector Interpolation (Standard)	$31.05 \pm 0.65\%$	$30.50 \pm 0.60\%$	$29.40 \pm 0.62\%$	-7.05%
Hybrid	$D_{train_SMOTE-IPF_feasible}$	SMOTE + Iterative Partitioning Filter	$32.80 \pm 0.58\%$	$31.10 \pm 0.52\%$	$30.65 \pm 0.55\%$	-5.80%
Hybrid	$D_{train_SMOTE-RkNN_feasible}$	SMOTE + Reverse k-NN Cleaning	$33.15 \pm 0.52\%$	$31.35 \pm 0.50\%$	$30.95 \pm 0.52\%$	-5.50%
Ours	$D_{train_BEDR_feasible}$	Boundary Entropy-Driven Resampling	$36.50 \pm 0.38\%$	$37.15 \pm 0.40\%$	$36.45 \pm 0.35\%$	-

call (Recall 37.15% vs. EDA 31.80%). The experimental results of RQ5 show that BEDR not only overcame the “Semantic Distortion” problem of traditional vector methods (SMOTE) but also broke through the “Knowledge Closure” limitation of traditional text methods (EDA), and achieved the best reshaping of the feature space through “Entropy-Driven Cleansing” superior to SMOTE-IPF/RkNN.

5. Conclusion and Future Work

5.1. Conclusion

This paper addresses the two core challenges of long-tail distribution and high semantic overlap in fine-grained TTP mapping tasks, identifying the failure of traditional resampling methods in zero-shot scenarios and the mode collapse issues inherent in naive LLM augmentation. To this end, we proposed the BEDR (Boundary Entropy-Driven Resampling) framework, establishing a new paradigm shift from “passive quantitative balancing” to “active quality optimization.”

Our core contributions and findings are summarized as follows:

- **Theoretical Innovation:** We formally defined Boundary Entropy (H_{bd}) as a metric to quantify the ambiguity of decision boundaries and theoretically demonstrated that minimizing

global H_{bd} is an effective pathway to reducing the upper bound of generalization error.

- **Methodological Breakthrough:** We constructed a Hybrid Vector Scaffold that combines structured intent with unstructured text, providing a robust reference system for entropy calculation. Simultaneously, the Structured Knowledge-Enhanced Prompting mechanism, based on action-chain extraction, ensures sample diversity and controllability at the source.
- **Experimental Validation:** Extensive experiments based on the MITRE ATT&CK dataset confirmed the superiority of BEDR. Addressing the cold start problem, BEDR successfully restored classification capabilities for 105 zero-shot categories, significantly outperforming baseline models. Regarding sample quality, BEDR achieved a higher Macro-F1 (32.76% vs. 28.72%) with fewer samples compared to the naive LLM strategy. Visualization analysis further confirmed that BEDR effectively overcame Mode Collapse, generating high-quality samples characterized by structured diversity (intra-cluster sparsity) and clear separability (large inter-cluster margin). In comparisons with related methods on the feasible subset ($N \geq 2$), BEDR outperformed advanced techniques such as EDA and SMOTE variants,

proving that in high-dimensional NLP spaces, semantic generation based on entropy filtering is superior to rule-based perturbation or geometric interpolation. In summary, BEDR provides a systematic solution for constructing high-quality, clear-boundary datasets in the field of cybersecurity.

5.2. Future Work

Although BEDR has achieved encouraging results, deeper exploration is warranted in the following directions:

- **From Single-Label to Multi-Label:** Currently, BEDR focuses on single-label multi-classification tasks. However, in real-world threat reports, a single sentence often implies multiple TTPs. Future work will extend boundary entropy theory to Multi-Label Scenarios, exploring how to define and optimize boundaries when samples belong to intersecting manifolds.
- **From Data-Centric to Model-Centric:** The BEDR framework proposed in this paper resolves the fundamental bottleneck of “scarcity of high-quality data” in threat intelligence mapping. Since the feature space has been reshaped and purified by BEDR, our next research focus will shift to the optimization of model architectures. We plan to fine-tune domain-specific language models (e.g., SecureBERT) using high-quality datasets constructed by BEDR. Through the synergy of “High-Quality Data + Strong Models,” we aim to achieve higher classification performance metrics to meet the practical demands of production environments.
- **Dynamic and Continual Learning:** Cyber threats evolve rapidly. We plan to integrate Continual Learning techniques with BEDR, enabling the model to adaptively generate samples for emerging TTPs. The goal is to learn new knowledge while avoiding catastrophic forgetting of old categories, thereby maintaining a real-time threat knowledge base.

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